

Mini RAG Pricing Policy Assistant

A simple Q&A Retrieval-Augmented Generation (RAG) application designed to support pricing policy inquiries, built using **Python**, **Streamlit**, and **Google Gemini**. It allows users to submit policy-related questions while it automatically retrieves the most relevant policy statements and uses an LLM to generate accurate and relevant responses.

This project demonstrates the core RAG workflow without external frameworks, making it easy to understand how retrieval and generation work together.

****Disclaimer:** All pricing policies included in this project are fictional and are provided for demonstration purposes only.

Workflow

The application follows a standard RAG pipeline:

1. The user enters a pricing-related question.
2. TF-IDF converts the knowledge base into searchable vectors.
3. Cosine similarity identifies the most relevant policy statements.
4. The retrieved policies are added to the LLM prompt.
5. Google Gemini generates an answer using only the retrieved context.
6. The application displays both the generated answer and the retrieved sources.

Installation

Clone the repository:

```
git clone https://github.com/YOUR_USERNAME/mini-rag-pricing-assistant.git
```

Install the required packages:

```
pip install -r requirements.txt
```

Configuration

Create a `.env` file in the project directory:

```
GEMINI_API_KEY=your_api_key_here
```

The `.env` file should **not** be committed to GitHub.

Run the Application

```
python -m streamlit run app.py
```

The application will be available at:

```
http://localhost:8501
```

<https://github.com/user-attachments/assets/c5dfab7d-63ef-4793-b0f4-85ec9db6e718>

Example Questions to Ask:

- What discount is allowed for enterprise customers?
- When is manager approval required?
- What pricing policies apply to healthcare customers?
- What should sales emphasize for education customers?